

SIDDHANT HIRAVE

Computer Engineering Student

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SUMMARY

Computer Engineering student with hands-on experience developing Arduino-based robotic systems through team and individual projects. Experienced in sensor integration, line detection, motor control, steering logic, PWM tuning, testing and troubleshooting. Seeking opportunities in Robotics, Automation and autonomous systems development.

TECHNICAL SKILLS

Programming: C/C++, Python, Arduino programming

Robotics & Automation: Arduino, QTR-8RC, ultrasonic sensors, motors, motor drivers, sensor calibration, steering logic, PWM

Engineering: System testing, debugging, troubleshooting, iterative development, component research

Tools: Arduino IDE, Tinkercad, Git, GitHub, Visual Studio Code, Onshape

PROJECTS

Line Following Robot *Robotics Club Learning Project | 8-member team | 2 weeks*

- Coordinated an 8-member team through system-level learning before transitioning into role-specific responsibilities based on team strengths.
- Programmed and tested QTR-8RC sensor interfacing, calibration and threshold tuning for line detection.
- Implemented motor control and left/right steering logic, adjusting PWM values and Arduino code during iterative testing.
- Tested motors and compared code versions while troubleshooting inconsistent motor-speed behaviour during development.

Arduino Ultrasonic Radar *Individual Project*

- Built an Arduino-based radar system with approximately 180° scanning using ultrasonic distance sensing and servo control.
- Implemented distance measurement, object-detection logic and serial data output for a computer-based radar display.
- Integrated ultrasonic sensing, servo control and Arduino programming into a working sensor-based system.

Ultrasonic Obstacle Detection System *Arduino / Tinkercad*

- Developed Arduino logic for ultrasonic-based front, left and right obstacle detection.
- Implemented decision-making logic for real-time responses and tested the program in Tinkercad.
- Refined Arduino code based on observed simulation behaviour and sensor readings.

Python Todo List *Individual Project*

- Developed a command-line Todo List application supporting task addition, editing, deletion and search.
- Implemented persistent storage using .txt file read/write operations so saved tasks could be loaded across sessions.
- Structured the application using functions, loops, conditional logic and menu-driven program flow.

EDUCATION

Smt. Kashibai Navale College of Engineering, Sinhgad Vadgaon, Pune

B.E. Computer Engineering | Savitribai Phule Pune University (SPPU)

3rd Semester | Expected Graduation: 2029 | **SGPA: 8.86 (Semester 1)**

ACTIVITIES

Technical Team Member — Robotics Club | 100+ member student club

- Contributed to technical sessions and hands-on robotics activities while supporting teammates with component research and technical work.
- Gained practical experience with Arduino-based sensors, motors and embedded programming through team projects.